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Data Mining applied to Knowledge Management

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Abstract

The digital age is responsible for the continuous growth of data volume. Everything related to Information and Communication Technologies contributes to the exponential growth of data, such as social networks, financial technology, and scientific data. Data Mining is a method of extracting data from systems such as Business Intelligence, Big Data, Data Warehouse, so that the patterns extracted during the process are considered information, knowledge, in relation to the theoretical concepts that support them. In this sense, these patterns should be presented as representations of knowledge. In the test case presented, we observed the various phases of the knowledge discovery process and analysed data using regression and classification models, to support companies in the knowledge management.

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Keywords: Data Mining; Business Intelligence; Data Warehouse; Knowledge Discovery; Knowledge Management; Decision Support Systems.

1. Introduction

Organisations should consider that, in a process of obtaining knowledge for decision-making, the management of information with quality is fundamental for the conscious systematisation of data, through patterns that can be mouldable to respond to the market demands, as well as designing patterns using Data Mining techniques, in relation

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to the concepts that underpin these patterns. In this sense, we intend to explore Data Mining techniques to create patterns that allow automating internal data quality processes.

With globalization, there is an increase in competitiveness that forces organizations to constantly change to manage the amount of data generated and that support decision making. Also, data represent an indispensable source to produce information and knowledge necessary [4][5][18]. In this context, Data Mining technologies require superior training and adequacy of the company's resources to analyze the information available and essential for any decision-making process [9][19]. The results obtained in the Data Mining phase allow the creation of new ideas and conceptual models for organizations [3][19].

The objective is to contribute with a Data Mining model, which allows the evaluation of data and leads to knowledge management, starting with the analysis and treatment of received data, developing dashboards for visualisation and analysis of the information, and to present examples of the creation of classification and regression models.

The paper is structured as follows: the next section frames all the important concepts of the development of the work, such as: Knowledge Discovery in Databases, Decision Support Systems, Data Mining and Business Intelligence; the third section proposes the methodological approach; the fourth section presents the practical component of the work from a test case, with the acquisition and preparation of the development data of the ETL process (Extract, Transform and Load), creation of the Cube "OLAP" (Online Analytical Processing), connection to Power BI, with the creation and analysis of dashboards, as well as the elaboration of classification and regression models in Data Mining; finally the last section presents the conclusions and final considerations.

2. State of the art

To identify Data Mining procedures relevant to knowledge management, a literature review was carried out, including: Scopus, ScienceDirect, Web of Science, Elsevier's, Google Scholar, as they bring together a very comprehensive set of references in the area. The following search expressions were defined: "Data Mining", "Data Science", "Business Intelligence", "ETL", "Integration Services", "Data Warehouse", "Analysis Services" and "OLAP".

Data Mining is used to analyse and exploit large amounts of data to present valuable information. The process of knowledge discovery in databases besides being iterative, it is also interactive since it requires the user's participation whenever decision making is required [17]. Thus, "Knowledge Discovery in Databases" is composed of five main phases (Fig. 1): Selection, Pre-Processing, Transformation, Data Mining and Evaluation/Interpretation.

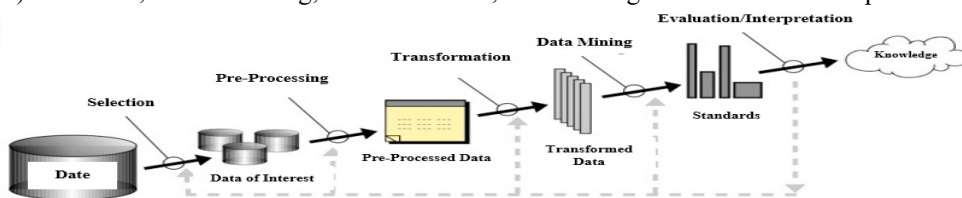


Fig. 1. Knowledge Discovery Process in Databases (adapted from Fayyad [6])

According to Oded et al [15], the five main stages are defined as: **Selection**: definition of objectives as well as the selection of data for the knowledge discovery process; **Pre-Processing**: improve data reliability, i.e., data are filtered and cleaned to make it consistent; **Transformation**: develop the best data models, to obtain quality information; **Data Mining**: definition of the choices according to the objectives defined in the previous steps; **Evaluation/Interpretation**: outline of the end of the knowledge discovery process in databases or readjust the patterns to repeat the process.

Decision Support Systems

A decision support system is an interactive application that combines data and mathematical models to solve problems regarding the management of organizations, as it can influence the transformation of information into useful knowledge in the decision-making process from systems such as Data Mining [12].

Data Mining

The phenomenon of data-driven discovery and decision making is the trend. Data Mining is the process of extracting knowledge from a large set of observed data, to define the unsuspected relationship and patterns in the data, presenting a novel, understandable and useful tool for users [1]. Data Mining is determinant for organisations, considering essential to have a deep analysis that allows the value of the pattern found to be defined, making them useful and relevant [11]. The objectives of Data Mining can be divided into two major categories, which are divided into subcategories according to the purpose of their use, as shown in Fig. 2.

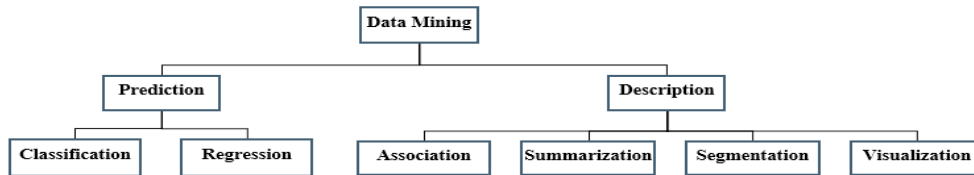


Fig. 2. Data Mining categories and subcategories (adapted from Pereira [16]).

Figure 2 identifies the Prediction category, consisting of Classification and Regression, as well as the Description category, consisting of Association, Summarization, Segmentation and Visualisation. Classification - is the process of assigning classes to objects according to their characteristics [16]. Regression: allows to set a value to a variable, such as predicting the probability of a person surviving a disease based on the results of diagnostic tests [17]; Association: determines the data that tend to compete in the same transaction, such as placing products together in a supermarket cart, market basket analysis [16]; Summarization: is the most evolved method, derived from summary rules, visualization techniques and discoveries of relationships between variables, applied to data analysis and automatic report generation [6]; Segmentation or Clustering: allows a heterogeneous division of a population into several homogeneous segments, with data grouped according to similarity [16]; Visualization: Data Mining results are presented through graphs and diagrams, allowing the representation of patterns and trends.

Business Intelligence (BI)

BI is a framework that combines architectures, data storage, analytical tools, computational applications, and methodologies that transform data into useful information to support decision-making [7][5][10][13][19]. It establishes the relationship between business, management, and technology, allowing the organization of indicators in real-time, so that employees, executives, and managers can make quality decisions [3][7][8][19].

BI systems improve the availability and quality of information [2][14], with support for Data Warehouses, OLAP systems for analytical data processing, and Data Mining mechanisms, for knowledge extraction, pattern identification, and predictive data models. According to Negash [13], the analysis of structured and unstructured data is transformed into decisions, as shown in Fig. 3.

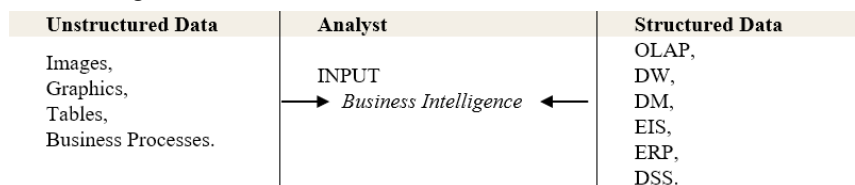


Fig. 3. Data division for *Business Intelligence* systems (adapted from Negash [13]).

3. Methodology

The methodology adopted in this article is Design Science Research (DSR). According to Hevner [9], this methodology, based on five phases, allows the production of innovative artifacts and responds to organizational problems. After its construction, the evaluation takes place. The objective is to provide new knowledge within the scope of scientific evidence. The key pieces to understanding the problem are the engineered artifacts. Through its construction and application, it is possible to know and understand the difficulty of the problem and define the solution.

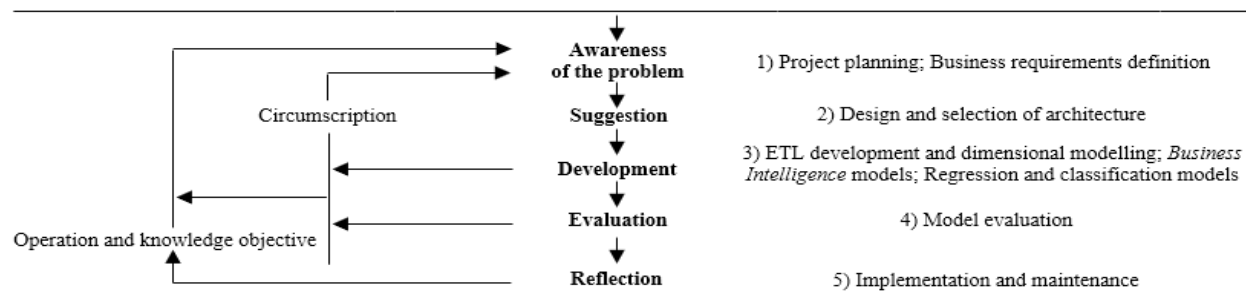


Fig. 4. Design Science Research methodology applied (adapted from Hevner [9]).

Fig. 4 describes the adaptation of the DSR methodology applied in this work. The first step consists of planning and defining objectives and requirements to ensure the development of the work. The second step is the integration of the various data storage technologies, in association with the design of the system architecture to meet the needs of the business. In the third phase, development with the transformation process (ETL) takes place. Dimensional modeling is instantiated in relational databases, multidimensional models and OLAP cubes, standardized reports developed in BI applications, as well as regression and classification models in Data Mining. The evaluation (fourth phase) of the model must be taken as a reference in the data analysis process and the presentation of the results, guaranteeing a good perception of them. The last phase, implementation, and maintenance, must ensure that all the initial objectives outlined are achieved, promoting acceptance of the business to support the decision-making process.

4. Solution Development

The main objective is to present an organizational knowledge discovery database. The database used to represent the solution was obtained from a scientific community opened platform, allowing regression and classification models to be created. Several tools like MS SQL Server, Visual Studio, MS Excel and Power BI were used to perform the development and implementation work.

Practical organization of tasks

Fig. 5 allows to observe the practical part, divided into two components: the acquisition of knowledge with the collection of data from the dataset in CSV format (Comma-separated values), organised in the MS Excel & Power Query environment (phase 1). This was followed by the design of the Data Warehouse architecture in MS SQL Server Management Studio. In phases 2, 3 and 4, the development of the solution took place, with the preparation of the data through the ETL process in Visual Studio - Integration Services Project, and data integration in the Data Warehouse, creating the favourable environment for the design of the analysis model, through Visual Studio - Multidimensional Project and Analysis Services Data Mining, report generator, OLAP Cube, the design in Data Mining of the decision tree algorithm and the accuracy model, within the scope of the regression and classification models. Finally, the use of Power BI, for visual analyses with the creation of dashboards or reports.

Knowledge acquisition		Development of the solution	
1	• Data collection and study • Problem classification • Solution design • Selection and aggregation of research data		• Data preparation • ETL Process
		2	• OLAP Cube creation
		3	• Introduction to Power BI
		4	• Dashboard creation process • Analysis of the developed dashboards • Development of regression and classification models

Fig. 5. Tasks of the project stages.

Collection and analysis of the organizational database

The database used includes a set of data with the aim of supporting the process of assimilation of the concepts associated with the discovery of knowledge in databases, as well as the various regression and classification techniques

and algorithms available for *Data Mining*. The concepts used with the database, serve as an example, to be explored in various organisational environments. The data obtained from the website of the National Health Service (SNS) in Portugal, result from the monthly monitoring of the number of surgical interventions in hospital care, for the different types: “Programmed”, “Ambulatory Programmed”, “Conventional Programmed and Urgent”, allowing the creation of classification and regression models, considering the algorithm in Data Mining.

Working environment

This topic discusses the technologies, tools and languages used, describing its objectives and how they fit in each phase of the work.

1. Data analysis by means of a proof of concept from Dataset and modelling of the Data Warehouse

The Dataset is composed of four tables, where a "Fact" table is responsible for the aggregation of the attributes and the other three "Dimension" tables refer to the attributes that support the "Fact" table. After defining the existing relationships in the Dataset, a multidimensional model was developed (Figure 6) that represents the Data Warehouse model. The "Dimension" tables, formed by "Region", "Entity" and "Period", identify the region of the hospital unit; the entity or health institution; and the temporal period of the movements by years and months. The "Fact" table aggregates the key attributes of the "Dimension" tables, as well as the movements, as shown in Fig. 6.

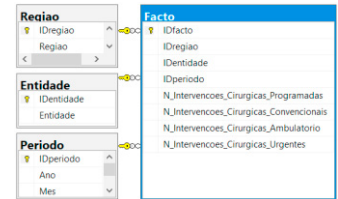


Fig. 6. Structure of Dimension and Fact tables of the Data Warehouse.

2. Design and Implementation of ETL processes (Dimensions and Facts)

The ETL process is responsible for extracting data from various sources and then transforming and loading the data into the target database through SQL Server Integration Services (SSIS) of Visual Studio Data Tools, as shown in Figure 7.

The development and implementation process took place as follows: Creation of the database in SQL Server; Extraction, Transformation and Loading of data (ETL), from the Dimension and Fact tables; Integration of data through the Data Source - Excel; Conversion of data to the Dimension and Fact tables; Lookup Match Output process, i.e., the corresponding output of the data in each of the tables until integration in the respective Dimension and Fact tables in the Data Warehouse, through OLE DB Destination, thus corresponding to the consolidation of the data in the Fact table. In other words, the data is normalized in the Dimension and Fact tables in the Data Warehouse for the creation of the OLAP Cube and regression and classification models (Fig. 7).

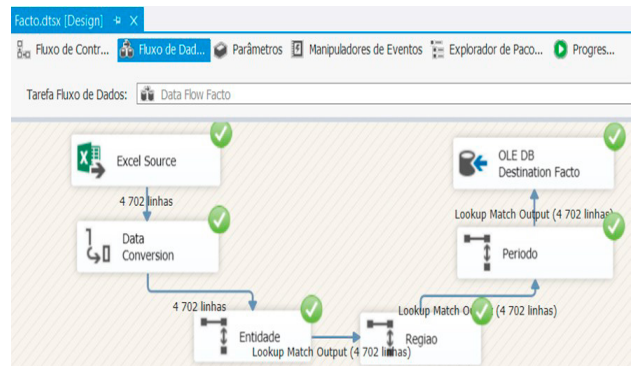


Fig. 7. Model developed in Visual Studio Data Tools for the ETL process.

3. Analysis Services' Multidimensional and Data Mining Project (OLAP Cube and Data Mining). OLAP Cube

The OLAP Cube allows data analysis through multidimensional views, identifies patterns and trends in the data. At this stage, the OLAP Cube was developed for subsequent data exploration through dashboards in the Power BI tool. The OLAP Cube was designed through the multidimensional model Analysis Services and respective connection to the Data Warehouse. The model used by the OLAP Cube allows, for example, from the attributes of the tables Fact, Region, Entity and Period, to perform a predictive analysis of the number of surgical interventions, organize and prepare the information in the context of Business Intelligence.

4. Visualisation Layer: Building Dashboards in Microsoft Power BI for the data sources

To complement the work done through the multidimensional model of the OLAP Cube and in an integrated way, the results were linked from the database of SQL Server Analysis Services to Power BI which is a tool for the visualization of data, as it allows the construction of tables and dynamic charts and filtering of large volumes of data, through a graphical and interactive interface. The dashboard presents, through the graphic representation, the number of surgeries performed by health region and by years, as shown in Fig. 8.

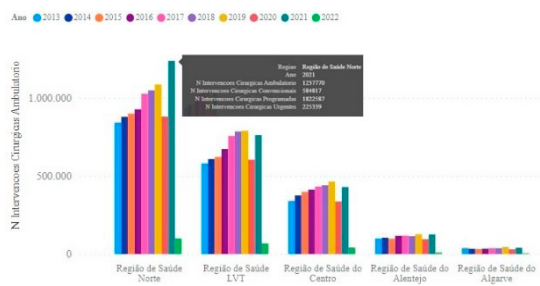


Fig. 8. Dashboards in Power BI.

Regression and classification models in Data Mining

The decision tree algorithm is a classification algorithm that works for predictive modelling. It supports prediction of discrete and continuous attributes. The structure of the algorithm used in the test model for Data Mining called Decision Tree, is formed by a set of elements, Node and Leaf, where the Node can lead to another Node or to a Leaf and the Leaf is itself. We can observe the Node as an attribute that exists in the dataset, and the Leaf as the class of the object.

The input data from the dataset refer to the result of the monthly monitoring of the number of scheduled surgical interventions in SNS hospital care. In the Decision Tree algorithm (regression and classification), we consider the data in a training set to create the Data Mining model and a test set to check the accuracy of the model, based on the percentage of data for testing and the maximum number of cases in the test dataset provided. In this regard, we considered as predictive element the region and input elements the period, the entity and the number of scheduled surgical interventions.

Figure 9 shows the Decision Tree with the results of the test case that we proposed to develop. We can conclude that for the number of cases used to train the model, representing 70% of the 4702 cases, the probability of the scheduled surgical interventions per region is the following: 35.19% in the Northern health region; 9.26% in the Alentejo health region; 2.61% in the Algarve health region; 19.93% in the Centre health region; and 33.02% in the Lisbon and Tagus Valley (LVT) health region. Also, as an example, we can observe the results presented in one of the leaves of the tree, for the entities whose code is greater than or equal to 190 and less than 195, with 39.18% probability of carrying out programmed surgical interventions in the North health region, 40.47% in the Alentejo health region, 0.05% in the Algarve health region, 20.26% in the Centre health region, and 0.05% in the LVT health region. In other words, the Nodes and leaves present the probabilities that characterise the regression and classification model, created based on the decision tree algorithm.

The accuracy graph shown in Fig. 9, represents almost 100% of the population correlation and the prediction probability, which represents the assertiveness of the Data Mining algorithm used for the work.

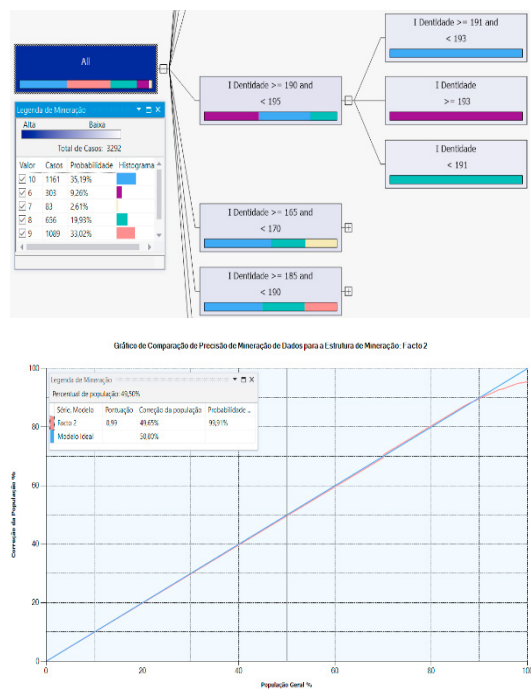


Fig. 9. Decision Tree Algorithm and Regression and Classification Model.

5. Discussion and conclusion

After the characterization of the various phases of the work, some conclusions and results can be considered. The first coincides with the importance of Data Mining to support data knowledge management for organizations. The data stored in various sources should be extracted and explored from systems that allow the conscious systematization of data, creating patterns that can be moldable to respond to the market, as well as designing patterns using Data Mining techniques, allowing the automation of internal quality processes. In this sense, the importance of the state of the art should be recognized, allowing the study and characterization of platforms, applications, and systems for the process of knowledge discovery in databases.

The creation of the artefact, based on DSR methodology, allowed the Business Intelligence system, supported by the regression and classification models in Data Mining, to find the plausible solution to the problem. After analysing the problem and according to this methodology, the solution found was to create a cataloguing typology, which allowed to identify how and which data extraction and exploration tools best meet user requirements, as well as to determine the regression and classification algorithm in Data Mining.

Thus, the development of the test case presents a new definition for data extraction and exploration activities. In addition, the development tools to be used in an interactive and multidimensional environment, allowing the extraction of knowledge from data for Business Intelligence, contribute to improve the decision process, in the scope of data science with Data Mining applied to knowledge management.

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